Chapter 8B Vertex-colorings

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The concept of vertex coloring of a graph can be used to model many scheduling problems, optimal assignment of channels to radio stations and optimal assignment of spectrum frequencies to mobile operations. However, the topic originated with the following "map coloring conjecture".

At most four colors are required to color any map of a country so that adjacent states receive different color.

This problem was first raised by Francis Guthrie in a letter to De Morgan in 1852. For a long time, it was a fascinating open problem among mathematicians. Arthur Cayley ($1821 - 1852$) and G.D. Birkhoff ($1884 - 1944$) made it known to a wider community by their presentations and innovative contributions. It was settled in affirmative by K. Appel, W. Haken and J. Kosh in 1977.

8B.1 Basic definitions

Definition. *Any function* $f : V(G) \to \{1, 2, \ldots, k\}$ *is called a* ***k-vertex-coloring*** *of a graph* G *if* $f(u) \neq f(v)$*, for any two adjacent vertices* $u, v \in G$*. The integers*

*$1, 2, \ldots, k$ are called the **colors**. Equivalently, a k-vertex-coloring of G is a partition $(V_1, \ldots, V_k)$ of $V(G)$ where each V_i is an independent set of vertices in G.*

Remarks.

- Every graph G admits a n-vertex-coloring. However, given k, G may not admit a k-vertex-coloring.
  Figure 8B.1 shows various colorings of $K_4 - e$.

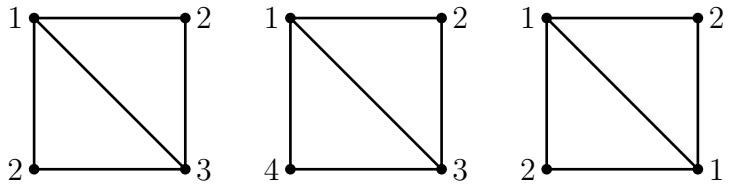


Figure 8B.1: (a) shows a 3-vertex-coloring, (b) shows a 4-vertex-coloring and (c) shows a coloring which is not a 2-vertex-coloring.

- If G admits a k-vertex-coloring, then it also admits a $(k+1)$-vertex-coloring.

The above remarks motivate the following important concept.

Definition. *The minimum integer k such that G admits a k-vertex-coloring is called the **vertex-chromatic-number** of G. It is denoted by $\chi(G)$ or $\chi_0(G)$. In other words, $\chi(G)$ is the minimum number of colors required to color the vertices such that no two adjacent vertices of G receive the same color.*

The following table shows the vertex-chromatic number of some standard graphs.

Remarks.

- If H is the underlying simple graph of G, then $\chi(H) = \chi(G)$.
- If $G_1, G_2, \ldots, G_t$ are the components of G, then

$$\chi(G) = \max\{\chi(G_1), \chi(G_2), \ldots, \chi(G_t)\}.$$

G	K_n	K_n^c	Bipartite graph	Petersen graph
$\chi(G)$	n	1	2	3

Table 8B.1: Vertex-chromatic number of some standard graphs.

- If $B_1, B_2, \ldots, B_p$ are the blocks of G, then

$$\chi(G) = \max\{\chi(B_1), \chi(B_2), \ldots, \chi(B_p)\}.$$

- In view of the above remarks, the study of vertex-colorings may be restricted to simple connected graphs.

Theorem 8B.1. *Every tree with two or more vertices is 2-chromatic.*

Proof. Select any vertex v in a given tree T. Consider T as a rooted tree at vertex v. Color the vertex v with 1. Color all vertices adjacent to v with color 2. Next color the vertices adjacent to these using color 1 (vertices at distance 2 from v). Continue this process till every vertex in T is colored. Now in T we find that all vertices at odd distances from v have color 2, while v and vertices at even distances from v have color 1. Now along any path in T the vertices are of alternating colors. Since there is one and only path between any two vertices in a tree, no two adjacent vertices have the same color. Thus T has been properly colored with two colors. □

Remark. *Even though a tree is 2-chromatic, not every 2-chromatic graph is a tree. For example, a cycle of even order is 2-chromatic.*

A simple characterization of all 2-chromatic graphs is given by König.

Theorem 8B.2. *A graph with at least one edge is 2-chromatic if and only if it has no cycle of odd length.*

Proof. Let G be a connected graph with cycles of only even length. Consider a spanning tree T in G. Using the coloring procedure as stated in the above theorem, let us properly color T with two colors. Now add the chords to T one by one. Since G has no cycles of odd length, the end vertices of every chord being replaced are differently colored in T. Thus G is colored with two colors, with no adjacent vertices having the same color. That is G is 2-chromatic.

Conversely, if G has a cycle of odd length, we would need at least three colors just for that cycle. Thus the theorem. □

8B.2 Cliques and chromatic number

Definition. *A subgraph Q of a graph G is called a **clique**, if any two vertices in Q are adjacent. A **maximal clique** is a clique that is not a subgraph of any other clique. The **clique number** of G is the integer $\omega(G) = \max\{|V(Q)| : Q \text{ is a clique in } G\}$.*

If H is a subgraph of G, then obviously $\chi(H) \leq \chi(G)$. Hence,

- For any graph G, $\chi(G) \geq \omega(G)$.

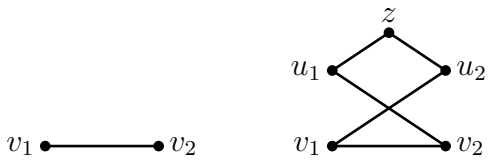
• Mycielski's theorem

There has been intense research to obtain upper bounds for $\chi(G)$ in terms of $\omega(G)$ alone. Questions like for which graphs $\chi(G) = \omega(G)$ or $\chi(G) = \omega(G) + 1$ or $\chi(G) \leq f(\omega(G))$ for a function f, have received wide attention. Mycielski (1955) showed that given any k, there exists a graph G with $\omega(G) = 2$ and $\chi(G) = k$. This result indicates that there is no upper bound for an arbitrary class of all graphs as a function of ω alone.

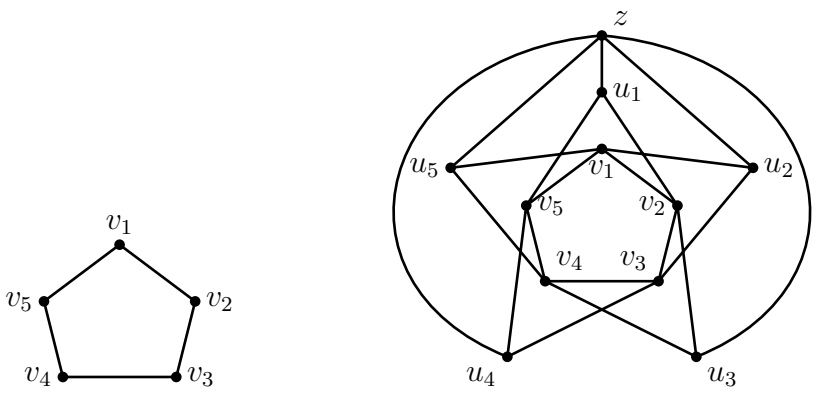
Theorem 8B.3. *For any given integer k (≥ 1), there exists a triangle-free graph G_k with vertex-chromatic number k. (Triangle-free $G := G$ has no K_3 as a subgraph.)*

Proof. G_k is constructed by induction on k. For $k = 1$ and 2, K_1 and K_2 are the required graphs. Next suppose that we have constructed a triangle-free graph G_k on p vertices $v_1, v_2, \ldots, v_p$. Construct G_{k+1} from G_k as follows.

1. Add $p + 1$ new vertices $u_1, u_2, \ldots, u_p$ and z to G_k.
2. Join u_i $(1 \leq i \leq p)$ to all those vertices to which v_i is adjacent in G_k.
3. Join z to every u_i.
   Figure 8B.2 shows this construction of G_3 from G_2 $(= K_2)$ and G_4 from G_3 $(= C_5)$.



(a) $G_2 \to G_3$



(b) $G_3 \to G_4$

Figure 8B.2: Mycielski's construction.

We shall prove that G_{k+1} is triangle-free and that $\chi(G_{k+1}) = k + 1$.

(a) G_{k+1} is triangle-free.

On the contrary, suppose that G_{k+1} contains a triangle $\mathcal{T}$. Since G_k is triangle-free, one of the vertices of $\mathcal{T}$ is z or u_i (for some i). If $z \in \mathcal{T}$, then the other two

vertices of $\mathcal{T}$ are u_g and u_h (for some g and h). This is a contradiction, since no two vertices in $\{u_1, \ldots, u_p\}$ are adjacent. If $u_i \in \mathcal{T}$ and $z \notin \mathcal{T}$, then the other two vertices of $\mathcal{T}$ are v_g and v_h (for some g and h). But then, v_i, v_g and v_h are the vertices of a triangle in G_k, which again is a contradiction. Hence, G_{k+1} is triangle-free.

(b) $\chi(G_{k+1}) = k + 1$.

Since, G_k is a subgraph of G_{k+1}, we have $\chi(G_{k+1}) \geq \chi(G_k) = k$. If $f : V(G_k) \to \{1, 2, \ldots, k\}$ is a k-coloring of G_k, then $g : V(G_{k+1}) \to \{1, 2, \ldots, k+1\}$ defined by

$$\begin{aligned} g(v_i) &= f(v_i), 1 \leq i \leq p, \\ g(u_i) &= f(v_i), 1 \leq i \leq p, \\ g(z) &= k + 1, \end{aligned}$$

is a $(k+1)$-coloring of G_{k+1}. Hence, $\chi(G_{k+1}) \leq k + 1$.

Next if possible suppose that $\chi(G_{k+1}) = k$, and let $h : V(G_{k+1}) \to \{1, 2, \ldots, k\}$ be a k-coloring of G_{k+1}. Without loss of generality, let $h(z) = k$. So, $h(u_i) \neq k$ for $i = 1, 2, \ldots, k$. Define $g : \{v_1, v_2, \ldots, v_p\} \to \{1, 2, \ldots, k-1\}$ by

$$g(v_i) = \begin{cases} h(v_i), & \text{if } h(v_i) \neq k \\ h(u_i), & \text{if } h(v_i) = k \end{cases}$$

It is easy to verify that g is a $(k-1)$-vertex-coloring of G_k. This is a contradiction to $\chi(G_k) = k$.

Hence, we conclude that $\chi(G_{k+1}) = k + 1$. □

8B.3 Greedy coloring algorithm

Finding $\chi(G)$ of an arbitrary graph G is a hard problem. So there have been attempts to use greedy and approximate algorithms and obtain tight bounds. However, it has been shown that there can be no approximate algorithm for finding $\chi(G)$ within a factor of $n^{1-\epsilon}$, for any $\epsilon > 0$, where n is the number of vertices in G. This deep result shows that finding the vertex-chromatic number of an arbitrary class of graphs is impossibly hard. Despite this negative result, greedy algorithm does use nearly χ colors if the input graph belongs to several interesting classes of graphs. This sections explores these aspects.

Greedy algorithm (Sequential algorithm/An example of an on-line algorithm.)

Input: A graph G and an ordering $(v_1, v_2, \ldots, v_n)$ of its vertices.

Output: A k-vertex-coloring of G, for some k.

Step 1: Color v_1 with 1.

Step 2: Having colored $v_1, v_2, \ldots, v_i$ with t colors, say $1, \ldots, t$, color v_{i+1} with the following policy. Let T be the set of colors used to color the vertices adjacent with v_i, then

$$color(v_{i+1}) = \begin{cases} \min(\{1, 2, \ldots, t\} - T), & \text{if } T \neq \{1, 2, \ldots, t\} \\ t + 1, & \text{if } T = \{1, 2, \ldots, t\}. \end{cases}$$

Step 3: Stop when v_n is colored.

An illustration:

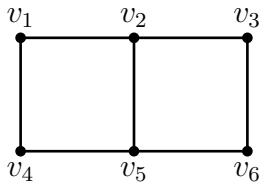
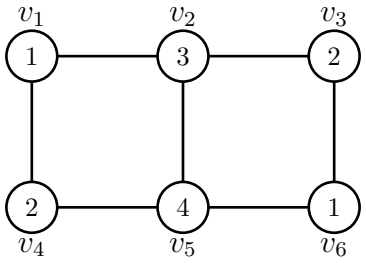


Figure 8B.3: A graph G.



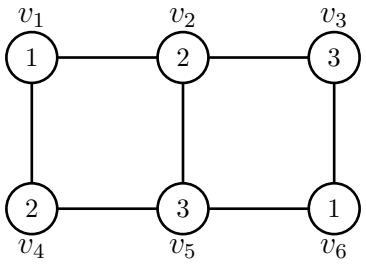
(a) An output of the greedy algorithm with input $(G; (v_1, v_6, v_3, v_4, v_2, v_5))$. The number inside the circle indicates its colors.

Remarks.

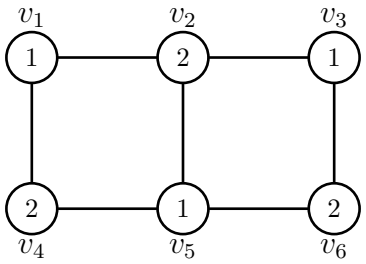
- Algorithm is called a greedy algorithm since we use the first available color at every step.
- It is called a sequential algorithm since the input is a sequence of the vertices and the vertices are colored sequentially.
- It is an example of an on-line algorithm since at every step we use the available data to color the vertex and do not change this color subsequently.
- The illustration indicates that the number of colors used by the algorithm depends on the ordering of the vertices.

Theorem 8B.4. *The greedy algorithm uses at most* $\Delta(G) + 1$ *colors.*

Proof. While coloring v_{i+1}, under Step 3, there are at most $\Delta(G)$ neighbours of v_{i+1} in $\{v_1, v_2, \ldots, v_i\}$, that is $|T| \leq \Delta(G)$. Hence, $color(v_{i+1}) \leq 1 + \Delta(G)$. This holds for every i. Hence, the result. □



(b) An output of the greedy algorithm with input $(G; (v_1, v_2, v_6, v_3, v_5, v_4))$.



(c) An output of the greedy algorithm with input $(G; (v_1, v_3, v_5, v_2, v_4, v_6))$. Since $\chi(G) = 2$, with this choice of vertex ordering the algorithm uses the minimum number of colors.

Figure 8B.4: Outputs of the greedy algorithm when three different vertex orderings are given as input.

Corollary. *For any graph G, $\chi(G) \leq 1 + \Delta(G)$.* □

Remark. If the vertices of G can be ordered $(v_1, v_2, \ldots, v_n)$ so that every v_i has at most $k-1$ neighbours in $\{v_1, v_2, \ldots, v_{i-1}\}$, then the greedy algorithm uses at most k colors.

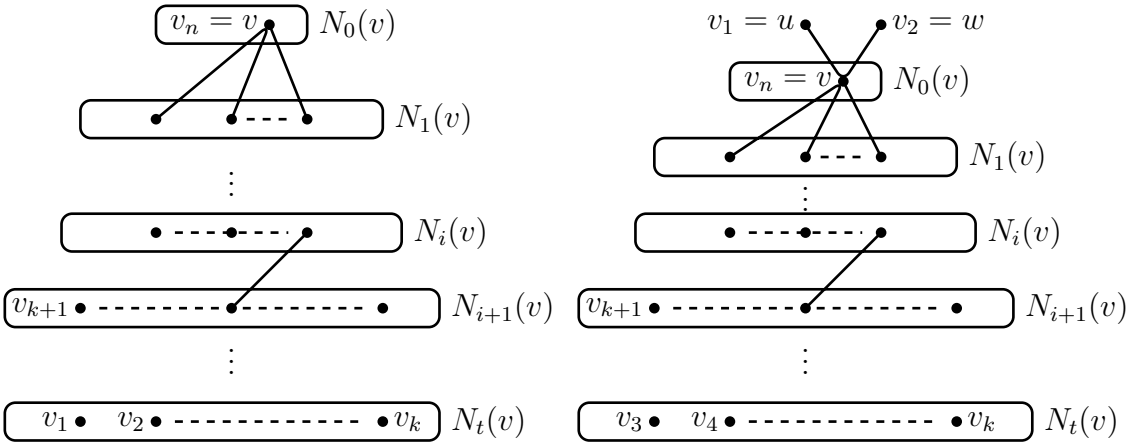
• Brooks theorem

Theorem 8B.5 (Brooks, 1941)**.** *If $G \notin \{C_{2p+1}, K_n\}$, then $\chi(G) \leq \Delta(G)$.*

Proof. (Lovasz, 1975) Without loss of generality, we assume that G is 2-connected. As remarked earlier, the idea of the proof is to get an ordering $(v_1, v_2, \ldots, v_n)$ of the vertices of G such that every v_i has at most $\Delta - 1$ neighbors in $(v_1, v_2, \ldots, v_{i-1})$.

Case 1: G is not regular, that is $\Delta(G) \neq \delta(G)$.

Let v be a vertex of minimum degree and let $t = \max\{d(v,x) : x \in V(G)\}$. Let $N_i(v) = \{x \in V(G) : d(v,x) = i\}$, for $i = 0, 1, \ldots, t$. Clearly, $V(G) = \cup_{i=0}^{t} N_i(v)$, and every $x \in N_i(v)$ is adjacent to some vertex in $N_{i-1}(v)$, for every i, $i = 1, \ldots, t$. Let $S = (N_t^+(v), N_{t-1}^+(v), \ldots, N_0(v))$ be an ordering of the vertices of G where $N_i^+(v)$ denotes a sequence of the vertices of $N_i(v)$ ordered arbitrarily; see Figure 8B.5a.



(a) Vertex ordering for Case 1 (b) Vertex ordering for Case 2

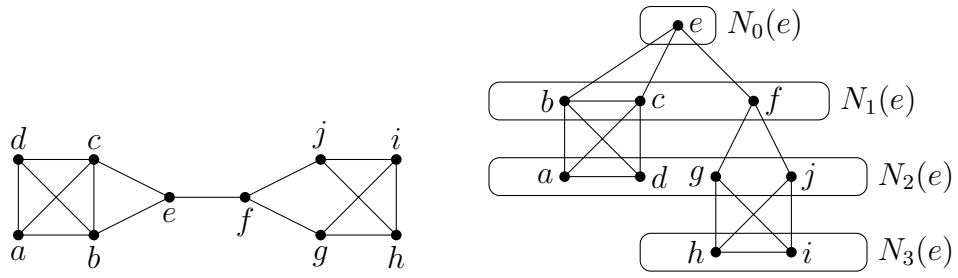
Figure 8B.5: An ordering of the vertices of G.

We apply greedy algorithm with S as the input. Every vertex $v_p(\neq v)$ in $N_{i+1}(v)$ is adjacent with a vertex in $N_i(v)$, $0 \leq i \leq t-1$. Hence when v_p is getting colored there are at most at most $\Delta - 1$ neighbors in $\{v_1, v_2, \ldots, v_{p-1}\}$. Therefore, v_p is colored with one of the colors in $\{1, 2, \ldots, \Delta\}$. Since $v_n(= v)$ has degree $\delta(G) < \Delta(G)$, v_n too has at most $\Delta - 1$ neighbors in $\{v_1, v_2, \ldots, v_{n-1}\}$. Therefore v_n is colored with one of the colors in $\{1, 2, \ldots, \Delta\}$.

We illustrate the above case with an example.

Illustration: Consider the graph shown in Figure 8B.6. The graph G is a non-regular graph with $\Delta(G) = 4$. By proceeding as in Case 1 with e as the initial vertex, the neighborhood decomposition of the graph yields an ordering of the vertices

(chosen arbitrarily within each level) given by $(h, i, a, d, g, j, b, c, f, e)$. Application of the greedy algorithm to this vertex ordering, gives a 4-coloring $(1, 2, 1, 2, 3, 3, 3, 4, 1, 2)$ to the graph as desired.



(a) Input Graph G (b) An ordering of G

Figure 8B.6: A graph G with $\Delta = 4$ with an ordering $(h, i, a, d, g, j, b, c, f, e)$ of the vertices.

Case 2: G is d-regular for some d; $2 \leq d \leq n-2$, since $G \neq K_n$.

If $d = 2$, then G is an even cycle, and so $\chi(G) = 2$. Next assume that $3 \leq d \leq n-2$. Note that the greedy algorithm uses at most Δ colors in this case too, if we can get an order $(v_1, v_2, \ldots, v_n)$ of the vertices such that every v_i $(\neq v_n)$ is adjacent with some vertex succeeding v_i and that v_n has at least two neighbors of the **same** color in $(v_1, v_2, \ldots, v_{n-1})$. Such an ordering is achieved by appealing to the following claim (whose proof is left as an exercise).

Claim: In any 2-connected k-regular graph $(3 \leq k \leq n-2)$, there exist three vertices u, v and w such that $(u, v), (v, w) \in E(G)$, $(u, w) \notin E(G)$ and $G - \{u, w\}$ is connected.

Let u, v, w be as in the claim and let $S = (v_1 = u, v_2 = w, N_t^+(v), \ldots, N_0^+(v))$, where $N_i(v) = \{x \in V(G - \{u, w\}) : d(x, v) = i\}$, $i = 0, 1, 2, \ldots, t$ and $N_i^+(v)$ is the set of vertices in $N_i(v)$ ordered arbitrarily; see Figure 8B.5b. With S as the input, the greedy algorithm colors both the vertices v_1 and v_2 with color 1. As in case 1, every vertex v_i $(3 \leq i \leq n-1)$ is adjacent with some vertex succeeding v_i in S. Hence

when v_i is getting colored it has at most $\Delta - 1$ colors appearing on its neighbors in $(v_1, v_2, \ldots, v_{i-1})$. Therefore, the greedy algorithm colors v_i with one of the colors in $\{1, 2, \ldots, \Delta\}$. Finally when v_n is getting colored it has at most $\Delta - 1$ colors appearing on its neighbors in $(v_1, v_2, \ldots, v_{n-1})$ since v_1 and v_2 are its neighbors and both are colored 1. Therefore, the greedy algorithm colors v_n too with one of the colors in $\{1, 2, \ldots, \Delta\}$. □

8B.4 Chromatic Partitioning

A proper coloring of a graph naturally induces a partitioning of the vertices into independent sets. For example, in Figure 8B.4(a) produces a partitioning of 4 independent sets $\{v_1, v_6\}, \{v_3, v_4\}, \{v_2\}, \{v_5\}$ while in Figure 8B.4(c) produces a partitioning of 2 maximal independent sets $\{v_1, v_3, v_5\}, \{v_2, v_4, v_6\}$.

Consider a χ-chromatic graph G of n vertices properly colored with χ different colors. Since the largest number of vertices in G with the same color cannot exceed the independence number $\alpha(G)$, we have the inequality

$$\chi(G) \geq \frac{n}{\alpha(G)}$$

.

- **Finding all maximal independent sets**

A reasonable (but not very efficient for large graphs) method for obtaining all maximal independent sets in any graph can be developed using Boolean arithmetic on the vertices. Let each vertex in the graph be treated as a Boolean variable. Let the logical (or Boolean) sum $a + b$ denote the operation of including vertex a or b or

both; let the logical multiplication ab denote the operation of including both vertices a and b, and let the Boolean complement a' denote that vertex a is not included.

For a given graph G we must find a maximal subset of vertices that does not include the two end vertices of any edge in G. Let us express an edge (x, y) as a Boolean product, xy, of its end vertices x and y, and let us sum all such products in G to get a Boolean expression

$$\phi = \sum xy, \text{ for all } x, y \in G.$$

Let us further take the Boolean complement rp' of this expression, and express it as a sum of Boolean products:

$$\phi' = f_1 + f_2 + \cdots + f_k.$$

A vertex set is a maximal independent set if and only if $\phi = 0$ (logically false), which is possible if and only if $\phi' = l$ (true), which is possible if and only if at least one $f_i = 1$, which is possible if and only if each vertex appearing in f_i (in complemented form) is excluded from the vertex set of G. Thus each f_i will yield a maximal independent set, and every maximal independent set will be produced by this method.

Consider the graph in Figure 8B.7,

$$\begin{aligned}
\phi &= ab + bc + cd + bd + de + ef + fg + eg \\
\phi' &= (a' + b')(b' + c')(c' + d')(b' + d')(d' + e')(e' + f')(f' + g')(e' + g') \\
\phi' &= b'e'f' + b'e'g' + a'c'd'e'g' + a'c'd'e'f' + b'c'd'f'g'.
\end{aligned}$$

we get upon simplification using the boolean arithmetic ($a+a=a, aa=a, a+ab=a$). Now if we exclude from the vertex set of G vertices appearing in any one of these

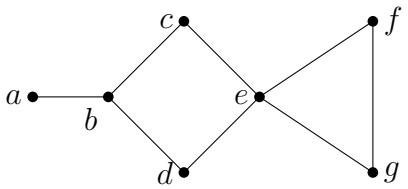


Figure 8B.7: A graph G

five terms, we get a maximal independent set. The five maximal independent sets are $acdf, acdg, bg, bf,$ and ae. These are all the maximal independent sets of the graph.

Finding Independence and Chromatic Numbers: Once all the maximal independent sets of G have been obtained, we find the size of the one with the largest number of vertices to get the independence number $\alpha(G)$. The independence number of the graph in Figure 8B.7 is four.

To find the chromatic number of G, we must find the minimum number of these (maximal independent) sets, which collectively include all the vertices of G. For the graph in Figure 8B.7, the sets $\{a, c, d, f\}, \{b, g\}$, and $\{a, e\}$, for example, satisfy this condition. Thus the graph is 3-chromatic.

Chromatic Partitioning: Given a simple, connected graph G, partition all vertices of G into the smallest possible number of disjoint, independent sets. This problem, known as the chromatic partitioning of graphs, is perhaps the most important problem in partitioning of graphs.

By enumerating all maximal independent sets and then selecting the smallest number of sets that include all vertices of the graph, we can solve this problem. The

following four are some chromatic partitions of the graph in Figure 8B.7,

$$\{(a,c,d,f),(b,g),(e)\}, \qquad \{(a,c,d,g),(b,f),(e)\},$$
$$\{(c,d,g),(b,f),(a,e)\}, \qquad \{(c,d,f),(b,f),(a,e)\}$$

However, this method of chromatic partitioning (requiring enumeration of all maximal independent sets) is inefficient and needs prohibitively large amounts of computer memory.